

Employee Attrition Analysis Report

Predictive Modeling of Workforce Turnover

A statistical and machine-learning assessment of workforce turnover drivers, using a synthetically generated employee dataset.

Dataset size	1,470 employee records
Features analyzed	11 workforce attributes
Models evaluated	Decision Tree, Logistic Regression
Report date	July 16, 2026

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1. Executive Summary

This report presents a statistical and machine-learning analysis of employee attrition using a dataset of 1,470 employee records spanning 11 workforce attributes, including demographics, compensation, tenure, satisfaction scores, and working conditions. The objective was to identify the factors most associated with employee turnover and to evaluate whether these factors can be used to predict attrition reliably.

The overall attrition rate in the dataset is 6.1% (90 departures out of 1,470 employees). Two classification models were trained and compared: a Decision Tree and a Logistic Regression model. The Decision Tree achieved an accuracy of 67.3% and an ROC AUC of 0.539, while the Logistic Regression model achieved an accuracy of 63.9% and an ROC AUC of 0.507.

An ROC AUC close to 0.50 indicates performance close to random chance, meaning that neither model in its current form provides strong discriminative power between employees who leave and those who stay. This finding is itself an important result: it suggests that the available features, at least in combination with these two model types, do not capture a strong, learnable attrition signal in this dataset. Section 7 discusses this limitation and its implications in more detail.

Key observations from the exploratory analysis include:

- Tenure shows the clearest gradient: attrition falls from 13.3% for employees with 0–2 years of service to 2.2% for employees with 25+ years.
- Job satisfaction is inversely related to attrition, ranging from 9.6% at the lowest satisfaction level to 4.9% at the third level.
- Department-level attrition rates are fairly close together, ranging from 5.3% to 6.9%, suggesting department alone is a weak differentiator.
- The Decision Tree's feature-importance ranking places YearsAtCompany, Age, MonthlyIncome, and DistanceFromHome as the top contributors to its predictions.

2. Introduction & Objectives

Employee attrition — the rate at which employees leave an organization — is a recurring concern for workforce planning, recruitment budgeting, and organizational continuity. Understanding which employee characteristics correlate with departure can help HR and management teams design targeted retention interventions rather than relying on broad, undifferentiated programs.

This report was prepared with three objectives:

- **Describe** the distribution of attrition across departments, tenure bands, satisfaction levels, and working conditions.
- **Model** attrition as a binary classification problem using two commonly used algorithms — a Decision Tree and a Logistic Regression — and compare their predictive performance.
- **Interpret** which features the models rely on most, and translate those findings into practical considerations for retention strategy.

The analysis was conducted using Python, with pandas and NumPy for data handling, scikit-learn for model training and evaluation, and matplotlib/seaborn for visualization. Full methodology is described in Section 3.

A note on the underlying data. The dataset used in this analysis was programmatically generated (not sourced from a real organization's HR system) using a fixed random seed, with attrition probability constructed as a function of overtime status, satisfaction scores, tenure, commute distance, and income. This makes it well suited for demonstrating an end-to-end analysis and modeling workflow, but the specific figures in this report should be read as illustrative of the method rather than as findings about any real workforce. Section 7 elaborates on this point.

3. Data Overview & Methodology

3.1 Dataset Description

The dataset contains 1,470 employee records with 11 predictor features plus the binary target variable, Attrition (1 = left the company, 0 = retained). The table below summarizes the fields used in the analysis.

Field	Type	Description
Age	Numeric	Employee age in years (18-60)
Gender	Categorical	Male / Female
Department	Categorical	Sales, R&D, HR, Finance, IT
MonthlyIncome	Numeric	Gross monthly income
YearsAtCompany	Numeric	Tenure in years (0-40)
YearsInCurrentRole	Numeric	Years in current role
YearsSinceLastPromotion	Numeric	Years since last promotion
OverTime	Categorical	Whether employee regularly works overtime
JobSatisfaction	Ordinal (1-4)	Self-reported job satisfaction
EnvironmentSatisfaction	Ordinal (1-4)	Self-reported environment satisfaction
DistanceFromHome	Numeric	Commute distance
Attrition	Binary target	1 = left, 0 = retained

3.2 Modeling Methodology

Categorical variables (Gender, Department, OverTime) were label-encoded. The data was split into training (1,176 records, 80%) and test (294 records, 20%) sets using stratified sampling to preserve the attrition ratio in both sets. Features were standardized (zero mean, unit variance) prior to training the Logistic Regression model; the Decision Tree was trained on unscaled features, since tree-based splits are not sensitive to feature scale.

Two models were trained, both using class-balanced weighting to compensate for the low base rate of attrition (roughly 6% of employees):

- **Decision Tree Classifier** — max depth 5, minimum 15 samples per leaf, balanced class weights.
- **Logistic Regression** — L2-regularized, balanced class weights, up to 1,000 iterations.

Model performance was evaluated on the held-out test set using accuracy, precision, recall, F1 score, and ROC AUC. Because attrition is a minority class, accuracy alone is not a reliable performance indicator — a model that always predicts "retained" would already be correct roughly 94% of the time without

identifying a single at-risk employee. For this reason, ROC AUC, precision, and recall are weighted more heavily in the interpretation of results (Section 5).

4. Exploratory Data Analysis

4.1 Overall Attrition Rate

Of the 1,470 employees in the dataset, 90 left the company and 1,380 were retained, for an overall attrition rate of 6.1%.

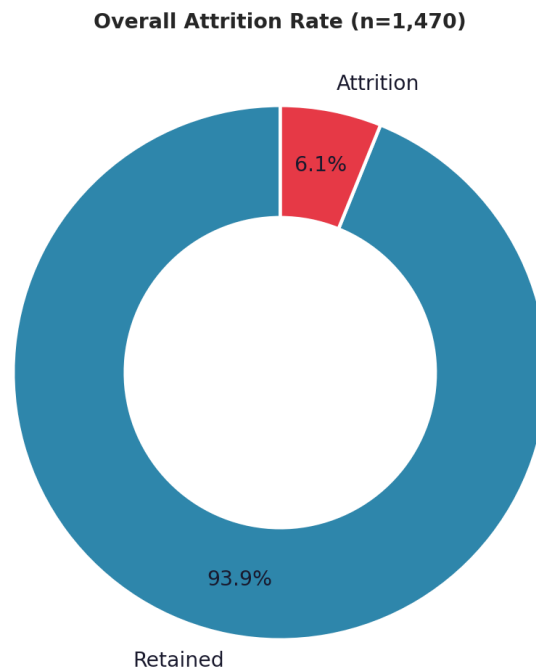


Figure 1. Overall attrition rate across all employees in the dataset.

4.2 Attrition by Department

Attrition rates by department range from 5.3% to 6.9%, a relatively narrow band around the overall average of 6.1%. IT and Sales sit slightly above average, while Finance and HR sit slightly below. The spread is not large enough to treat department as a strong standalone predictor.

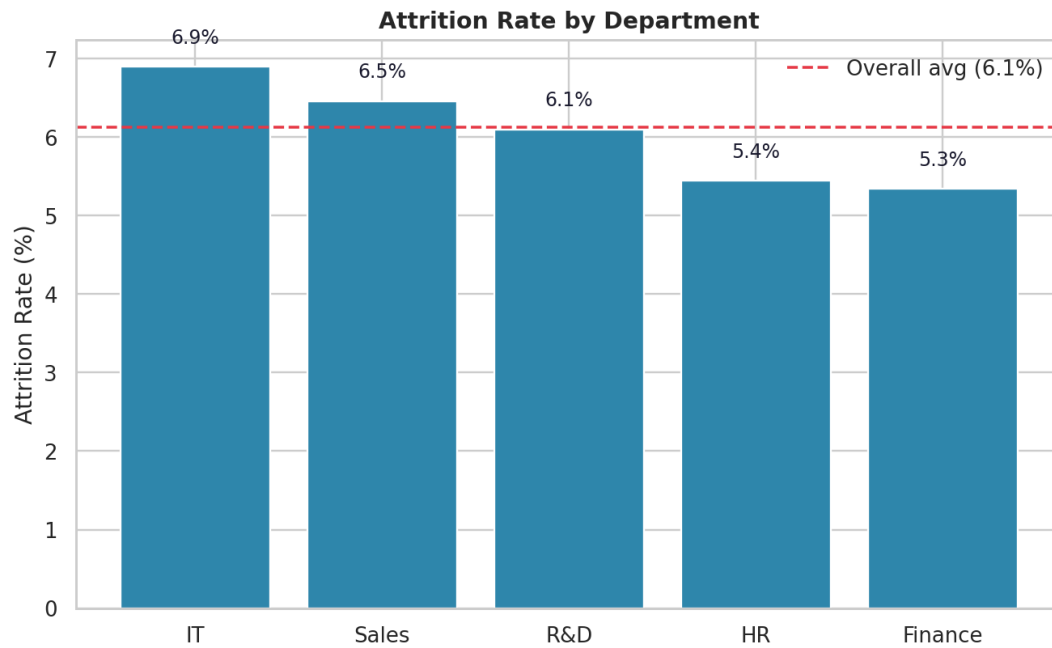


Figure 2. Attrition rate by department, with the dashed line marking the company-wide average.

4.3 Overtime & Job Satisfaction

Employees who regularly work overtime show an attrition rate of 6.2%, nearly identical to the 6.1% rate among those who do not. In this dataset, overtime status alone does not separate leavers from stayers as clearly as might be expected. Job satisfaction shows a clearer, though modest, pattern: employees at the lowest satisfaction level (score 1) have an attrition rate of 9.6%, compared with 4.9% at the middle-high level (score 3).

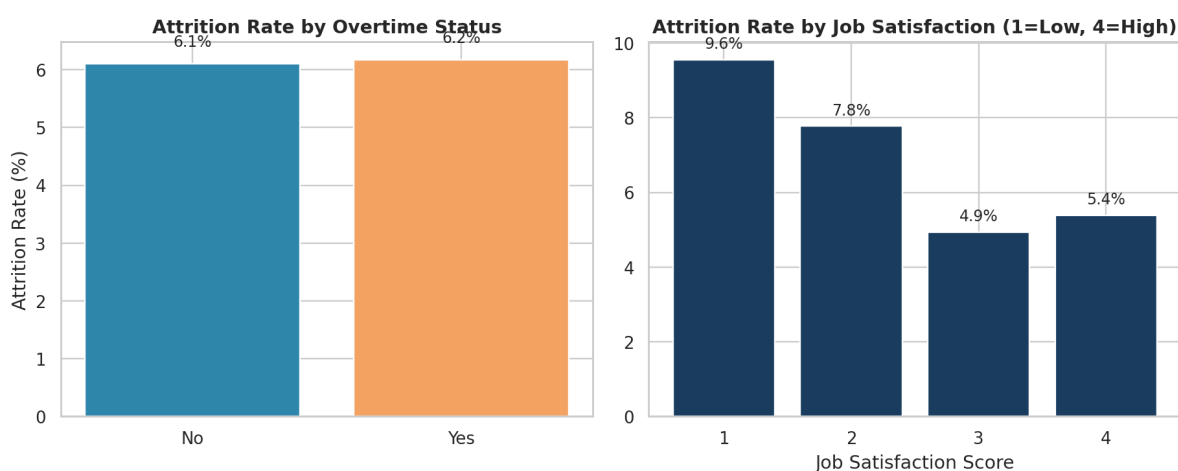


Figure 3. Attrition rate by overtime status (left) and job satisfaction score (right).

4.4 Income Distribution

Average monthly income across the workforce is 6,532, with a median of 6,446. The density plot below compares the income distributions of employees who left versus those who stayed; the two distributions overlap substantially, indicating income alone is a weak separator of attrition status in this dataset.

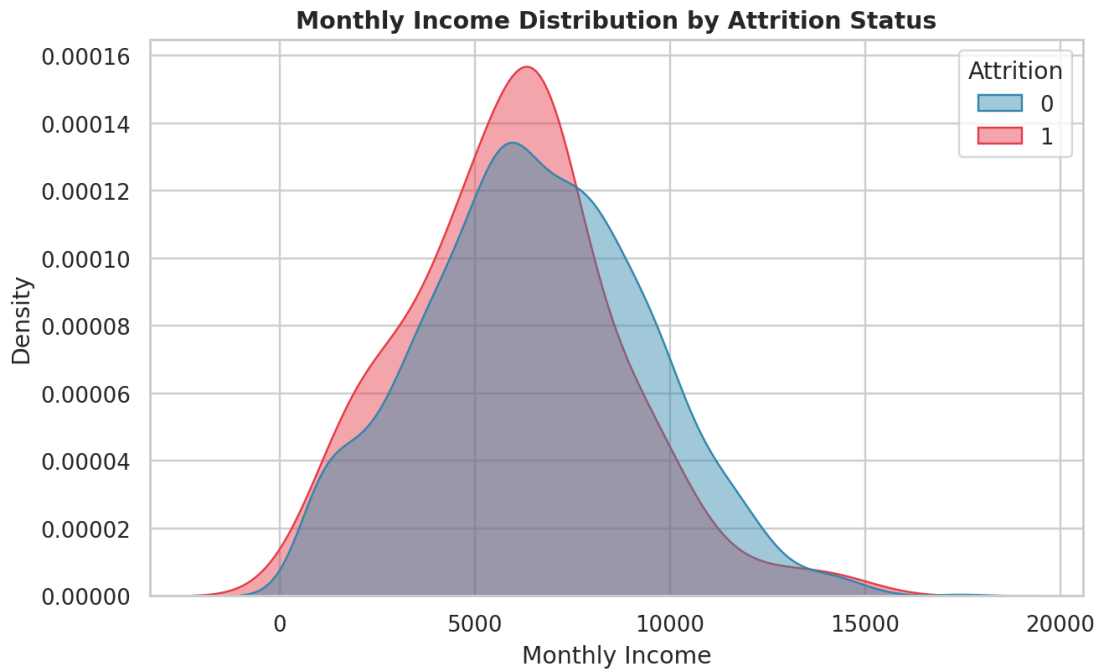


Figure 4. Monthly income distribution, split by attrition status (0 = retained, 1 = attrition).

4.5 Tenure & Attrition

Tenure produces the most visible gradient in the exploratory analysis. Attrition rate declines fairly steadily from 13.3% among employees with 0–2 years of service to 2.2% among employees with 25 or more years, consistent with newer employees being more likely to leave than long-tenured ones.

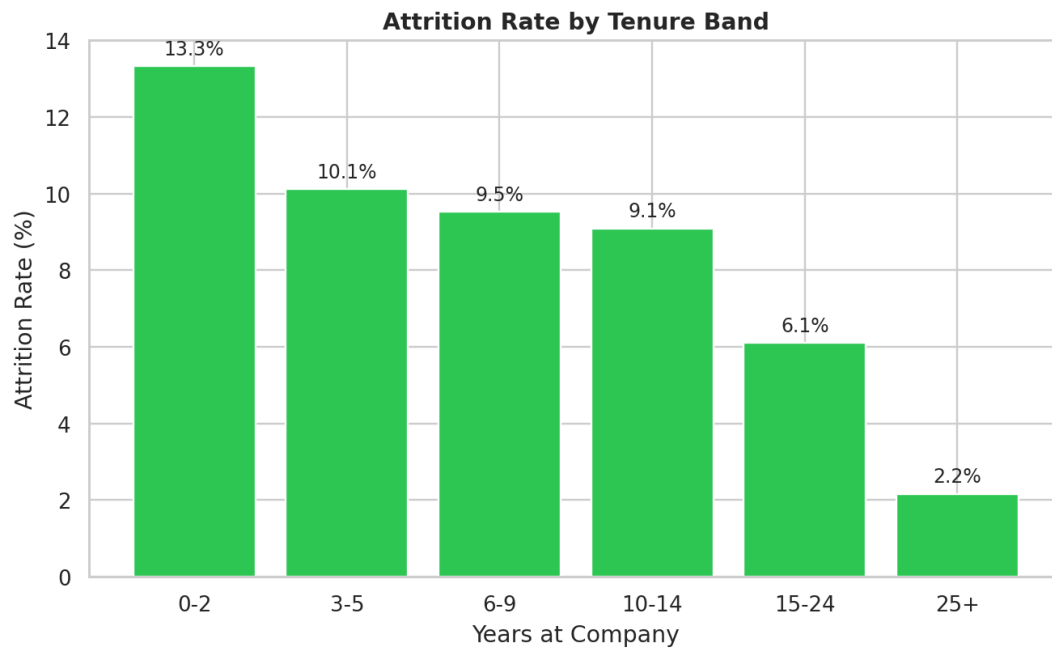


Figure 5. Attrition rate by tenure band.

4.6 Correlation Analysis

The correlation matrix below summarizes linear relationships among the numeric features and the attrition target. As is typical for attrition-style data, most individual correlations with the target are modest in magnitude, reinforcing that attrition here is driven by a combination of weak signals rather than any single dominant factor.

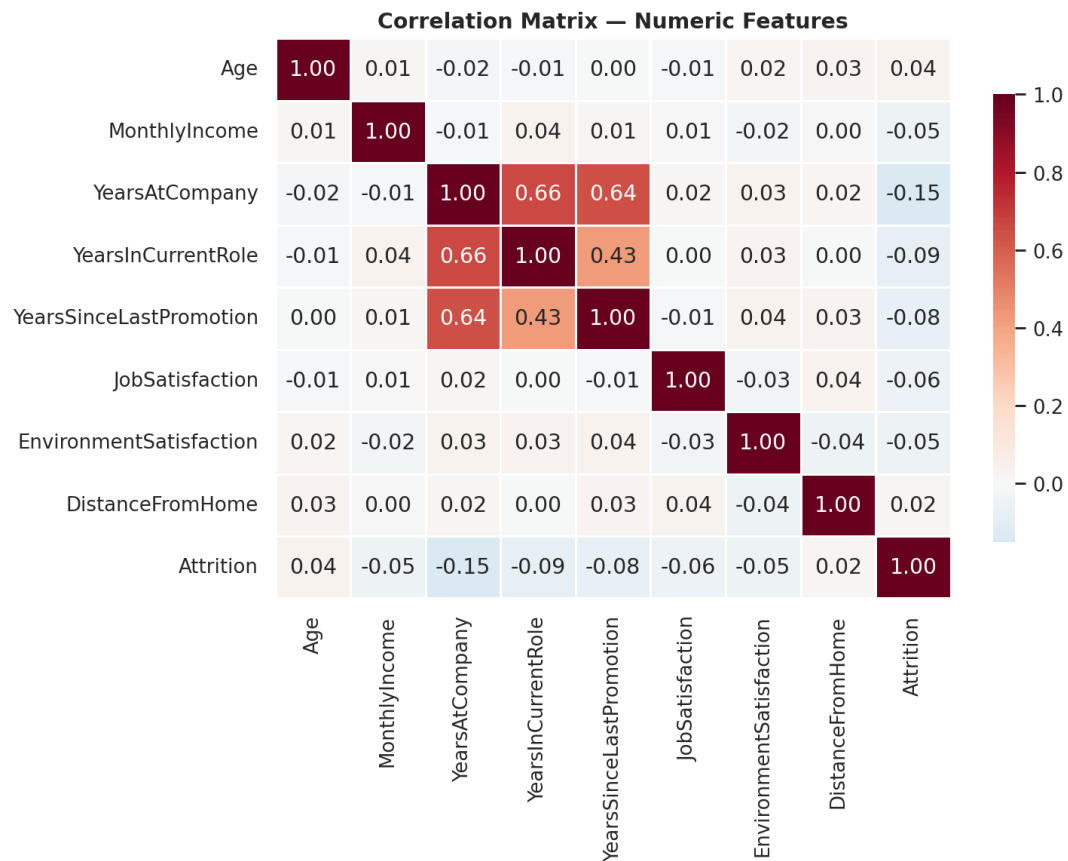


Figure 6. Correlation matrix of numeric features, including the Attrition target.

5. Predictive Modeling

5.1 Model Configuration

As described in Section 3.2, a Decision Tree and a Logistic Regression model were each trained on the same 80/20 stratified train-test split, with class balancing applied to account for the low base rate of attrition.

5.2 Performance Metrics

Metric	Decision Tree	Logistic Regression
Accuracy	67.3%	63.9%
Precision	5.7%	6.0%
Recall	27.8%	33.3%
F1 Score	9.4%	10.2%
ROC AUC	0.539	0.507

Both models show high accuracy but low precision, a direct consequence of the class imbalance: since roughly 94% of employees are retained, a model can score well on accuracy while still struggling to correctly flag employees who actually leave. Precision for both models sits near 6%, meaning that among employees flagged as at-risk, only a small fraction actually left in the test set. ROC AUC for both models is close to 0.50, indicating limited ability to rank at-risk employees above low-risk ones better than chance.

5.3 ROC Curve Comparison

The ROC curve plots the true positive rate against the false positive rate across all classification thresholds. A curve that closely tracks the diagonal (the dashed reference line) indicates performance close to random guessing.

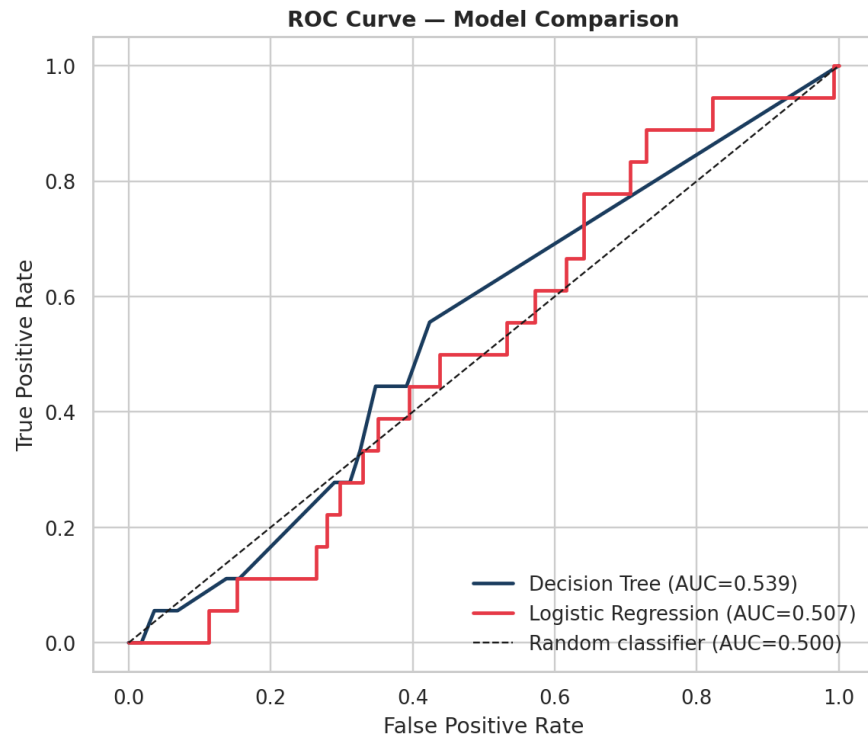


Figure 7. ROC curves for the Decision Tree and Logistic Regression models, with the diagonal representing a random classifier.

5.4 Confusion Matrices

The confusion matrices below show, for each model, how test-set predictions broke down across the four possible outcomes: true negatives, false positives, false negatives, and true positives.

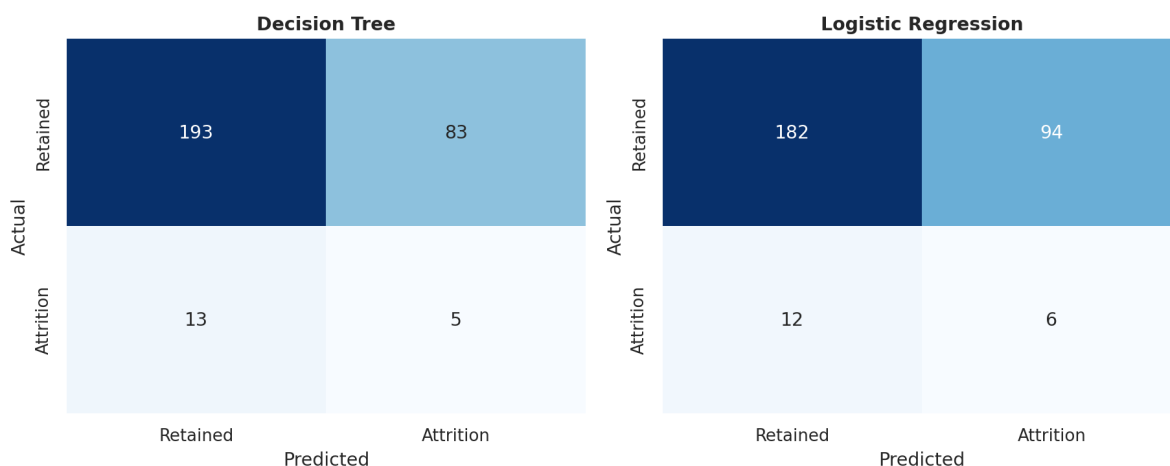


Figure 8. Confusion matrices for the Decision Tree (left) and Logistic Regression (right) models on the test set.

5.5 Feature Importance

For the Decision Tree, feature importance reflects how much each variable contributed to reducing impurity across all splits in the tree. YearsAtCompany dominates the ranking, followed by Age,

MonthlyIncome, and DistanceFromHome. JobSatisfaction, OverTime, and Department received negligible importance in this particular tree, despite being used in the process that generated the underlying attrition probabilities — a reminder that a single decision tree's importances depend heavily on which correlated features happen to be selected first during tree construction.

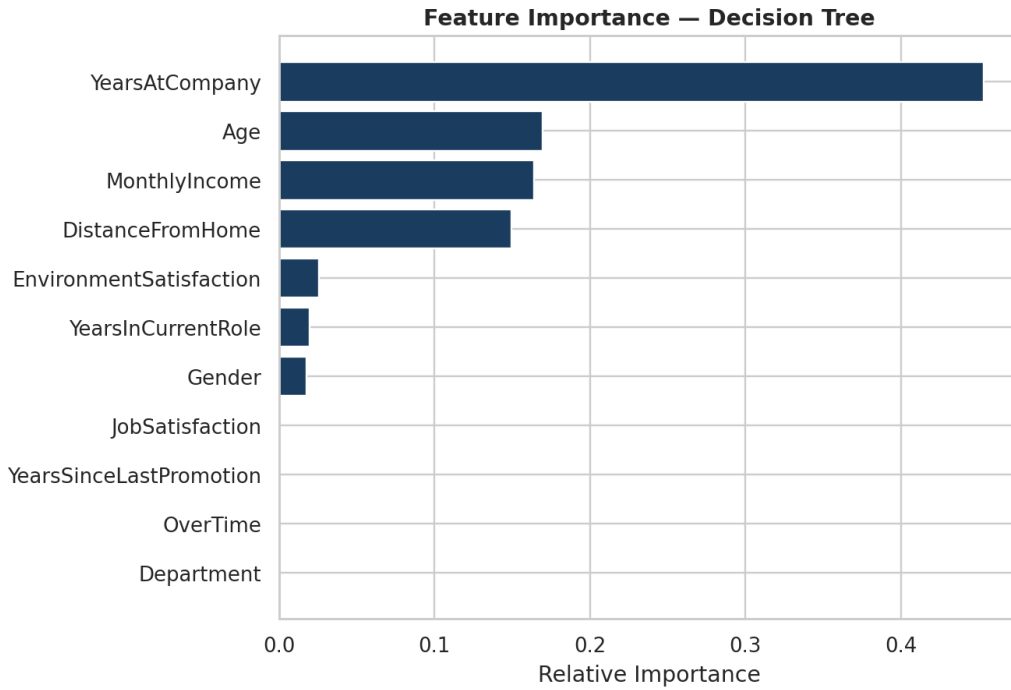


Figure 9. Relative feature importance from the Decision Tree model.

5.6 Decision Tree Visualization

The diagram below shows the top three levels of the trained Decision Tree, illustrating the sequence of splits the model uses to classify employees as likely to stay or leave.

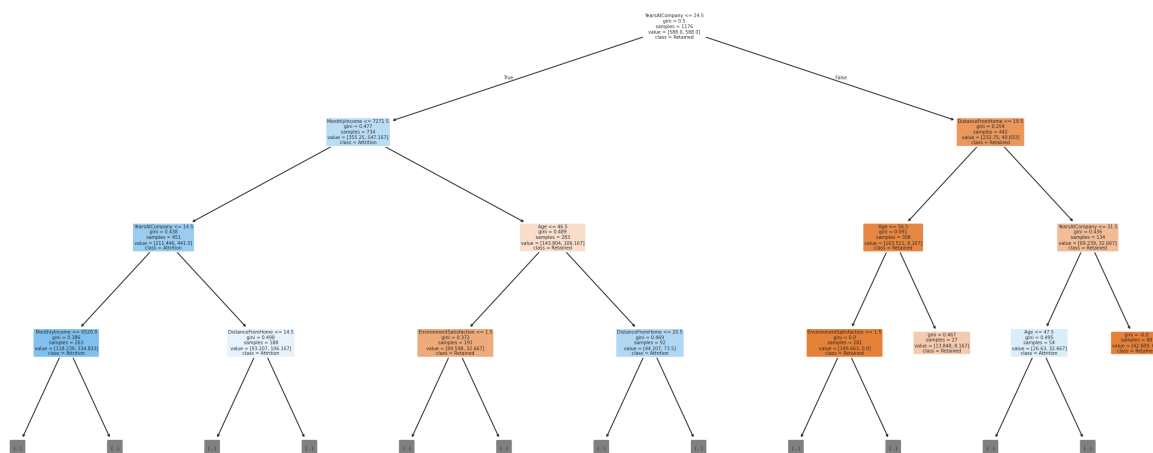


Figure 10. Decision Tree structure (top 3 levels shown).

6. Key Findings

- **Tenure is the strongest observed signal.** Attrition risk is markedly higher among newer employees and declines steadily with years of service.
- **Satisfaction matters, but modestly.** Lower job satisfaction correlates with higher attrition, though the effect size is smaller than tenure.
- **Department and overtime show limited standalone predictive value** in this dataset — differences across categories are small relative to overall variability.
- **Both models perform close to chance on ROC AUC** (0.50-0.54), indicating that, as configured, neither reliably separates likely leavers from likely stayers.
- **Class imbalance is a major driver of the low precision scores** observed for both models; with only ~6% of employees leaving, false positives naturally outnumber true positives at most thresholds.

7. Limitations

Synthetic data. The dataset analyzed here was generated programmatically rather than drawn from a real HR system. While it was constructed to loosely mimic realistic attrition dynamics, the specific relationships and magnitudes reported in this document should not be interpreted as findings about an actual workforce. This report should be read as a demonstration of the analysis and modeling pipeline.

Model performance. ROC AUC values near 0.50 indicate the current models have limited discriminative power on this data. This may reflect a genuinely weak signal-to-noise ratio in the underlying data-generating process, insufficient feature engineering, or the simplicity of the two model types used (a shallow decision tree and a linear logistic regression). More flexible models, additional features (e.g., manager relationship quality, recent performance ratings, engagement survey trends), or a larger sample size could plausibly improve predictive performance.

Class imbalance. With attrition present in only about 6% of records, precision for a positive prediction is inherently constrained; threshold tuning, resampling techniques (e.g., SMOTE), or cost-sensitive learning were not explored here and could improve the operational usefulness of a flagging model.

Single train/test split. Results are based on one 80/20 split rather than cross-validated estimates, so reported metrics carry some sampling variability.

8. Recommendations

- **Prioritize early-tenure retention.** Given the clear tenure gradient, onboarding quality, early mentorship, and first-year check-ins are reasonable areas to invest in further, pending validation on real data.
- **Track satisfaction trends, not just snapshots.** Since satisfaction shows a measurable relationship with attrition, monitoring changes in satisfaction over time may be more informative than a single survey point.
- **Expand the feature set before deploying a predictive model in production.** Additional signals such as engagement survey results, manager-relationship indicators, performance trajectory, and compensation-to-market benchmarks would likely improve model performance beyond the near-chance levels observed here.
- **Address class imbalance explicitly** if a production model is built — through resampling, class weighting (already applied here), or threshold calibration tied to a specific business cost of false positives versus false negatives.
- **Re-run this analysis on real HR data** once available, using the same pipeline, to validate whether the tenure and satisfaction patterns observed here hold in practice.

9. Conclusion

This analysis walked through a complete attrition-modeling workflow: data preparation, exploratory analysis, model training, and evaluation. Of the patterns examined, tenure and job satisfaction showed the most consistent relationships with attrition, while department and overtime status showed little standalone signal. The Decision Tree and Logistic Regression models achieved ROC AUC scores of 0.539 and 0.507 respectively — close to the 0.50 baseline of a random classifier — indicating that, in their current form, they are not yet reliable tools for flagging individual at-risk employees. The exploratory findings on tenure and satisfaction nonetheless offer a reasonable starting point for retention discussions, and the modeling pipeline established here provides a foundation that can be strengthened with richer data and further model development.